

Advanced Algebra

ANSWER KEY



Simplifying rational expressions

- 1 Simplify $\frac{x^2 - 4}{x^2 + x - 6}$.
$$\frac{x^2 - 4}{x^2 + x - 6} = \frac{(x-2)(x+2)}{(x-2)(x+3)} = \frac{x+2}{x+3}, x \neq 2.$$
- 2 Simplify $\frac{x^2 - 5x + 6}{x^2 - 9}$.
$$\frac{x^2 - 5x + 6}{x^2 - 9} = \frac{(x-2)(x-3)}{(x-3)(x+3)} = \frac{x-2}{x+3}, x \neq 3.$$
- 3 Combine into a single fraction: $\frac{2}{x+1} + \frac{3}{x-1}$.
$$\frac{2}{x+1} + \frac{3}{x-1} = \frac{2(x-1) + 3(x+1)}{(x+1)(x-1)} = \frac{2x-2+3x+3}{x^2-1} = \frac{5x+1}{x^2-1}.$$
- 4 Simplify $\frac{3x^2 - 12}{x^2 - 4x + 4}$.
$$\frac{3x^2 - 12}{x^2 - 4x + 4} = \frac{3(x-2)(x+2)}{(x-2)^2} = \frac{3(x+2)}{x-2}, x \neq 2.$$

Absolute value equations

- 5 Solve $|2x - 5| = 9$.
 $2x - 5 = 9 \Rightarrow x = 7$. $2x - 5 = -9 \Rightarrow x = -2$. Solutions: $x = 7$ or $x = -2$.
- 6 Solve $|3x + 1| = |x - 5|$.
Case 1: $3x + 1 = x - 5 \Rightarrow 2x = -6 \Rightarrow x = -3$. Case 2: $3x + 1 = -(x - 5) \Rightarrow 4x = 4 \Rightarrow x = 1$. Both check: solutions $x = -3$ or $x = 1$.
- 7 Solve $|x - 4| + 3 = 10$.
 $|x - 4| = 7$, so $x - 4 = 7 \Rightarrow x = 11$, or $x - 4 = -7 \Rightarrow x = -3$. Solutions: $x = 11$ or $x = -3$.
- 8 Solve $|4x + 3| = 11$.
 $4x + 3 = 11 \Rightarrow x = 2$. $4x + 3 = -11 \Rightarrow x = -3.5$. Solutions: $x = 2$ or $x = -3.5$.

Polynomial division, remainder & factor theorems

- 9 Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.
 $f(2) = 8 + 8 - 10 + 1 = 7$. The remainder is 7.

10 Given $f(x) = x^3 + 3x^2 - 4x - 12$:

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

$f(-3) = (-27) + 3(9) - 4(-3) - 12 = -27 + 27 + 12 - 12 = 0$, so by the factor theorem $(x + 3)$ is a factor.

b Hence factor $f(x)$ completely. Dividing: $f(x) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

11 Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.

Quotient $= x^2 + x - 2$, remainder $= 0$. So

$$x^3 - 2x^2 - 5x + 6 = (x - 3)(x^2 + x - 2) = (x - 3)(x + 2)(x - 1).$$

Logarithmic equations

12 Solve $\log_3(x + 5) = 2$.

$$x + 5 = 3^2 = 9 \Rightarrow x = 4.$$

13 Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.

$\log_2[(x + 1)(x - 1)] = 3 \Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$. Since $x > 1$ is required for $\log_2(x - 1)$ to be defined, reject $x = -3$. Solution: $x = 3$.

14 Solve $2\log_5 x = \log_5 16$.

$$\log_5(x^2) = \log_5 16 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4. \text{ Since } x > 0 \text{ is required, reject } x = -4. \text{ Solution: } x = 4.$$

15 Solve $\log_3(x - 2) = 4$.

$$x - 2 = 3^4 = 81 \Rightarrow x = 83.$$

Exponential equations

16 Solve $3^{2x-1} = 27$.

$$27 = 3^3, \text{ so } 2x - 1 = 3 \Rightarrow x = 2.$$

17 Solve $5^{x+2} = 5^{3x-4}$.

$$x + 2 = 3x - 4 \Rightarrow 6 = 2x \Rightarrow x = 3.$$

18 Solve $2^x = 15$ for x , giving your answer to three decimal places.

$$x = \frac{\ln 15}{\ln 2} \approx \frac{2.708}{0.693} \approx 3.907.$$

19 Solve $4^x = 2^{x+3}$.

Rewrite $4^x = 2^{2x}$, so $2x = x + 3 \Rightarrow x = 3$.

Nonlinear systems of equations

20 Solve the system $y = x^2 - 1$ and $y = 2x + 2$.

$x^2 - 1 = 2x + 2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0 \Rightarrow x = 3$ or $x = -1$. Solutions: $(3, 8)$ and $(-1, 0)$.

21 Solve the system $x^2 + y^2 = 25$ and $y = x + 1$.

Substitute: $x^2 + (x + 1)^2 = 25 \Rightarrow 2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0 \Rightarrow x = -4$ or $x = 3$. Solutions: $(-4, -3)$ and $(3, 4)$.

22 Solve the system $y = x^2 - 4x + 3$ and $y = -x + 3$.

$x^2 - 4x + 3 = -x + 3 \Rightarrow x^2 - 3x = 0 \Rightarrow x(x - 3) = 0 \Rightarrow x = 0$ or $x = 3$. Solutions: $(0, 3)$ and $(3, 0)$.

Rational equations and extraneous solutions

23 Solve $\frac{4}{x-1} = \frac{2}{x+1}$.

Cross-multiplying: $4(x + 1) = 2(x - 1) \Rightarrow 4x + 4 = 2x - 2 \Rightarrow 2x = -6 \Rightarrow x = -3$. Valid, since $x \neq \pm 1$.

24 Solve $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

Multiply through by $(x - 2)(x + 2)$:

$x(x + 2) + 2(x - 2) = 8 \Rightarrow x^2 + 4x - 4 = 8 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x + 6)(x - 2) = 0 \Rightarrow x = -6$ or $x = 2$.

Since $x = 2$ makes the denominator zero, it is extraneous. Solution: $x = -6$.

25 Solve $\frac{3}{x+4} = \frac{1}{x-2}$.

Cross-multiplying: $3(x - 2) = x + 4 \Rightarrow 3x - 6 = x + 4 \Rightarrow 2x = 10 \Rightarrow x = 5$. Valid, since $x \neq -4, 2$.

Systems of inequalities and linear programming

26 A feasible region is bounded by the constraints $x \geq 0$, $y \geq 0$, $x + y \leq 6$, and $2x + y \leq 10$. Find the vertices of the feasible region.

Intersecting boundary lines pairwise (and checking each candidate point satisfies every constraint) gives vertices $(0, 0)$, $(5, 0)$, $(4, 2)$, and $(0, 6)$.

- 27 Using the feasible region from the previous question, find the maximum value of $P = 3x + 2y$ over the region, and state where it occurs.

Evaluating P at each vertex: $(0, 0) \rightarrow 0$; $(5, 0) \rightarrow 15$; $(4, 2) \rightarrow 16$; $(0, 6) \rightarrow 12$. The maximum is $P = 16$, occurring at $(4, 2)$.

A well-designed word problem: production planning at a furniture workshop

- 28 This question has three parts. A furniture workshop makes tables and chairs. Each table requires 4 hours of carpentry and 2 hours of finishing; each chair requires 1 hour of carpentry and 1 hour of finishing. Each week the shop has at most 32 hours of carpentry time and 20 hours of finishing time available. Each table earns a profit of 70 dollars and each chair earns a profit of 30 dollars. Let x be the number of tables and y the number of chairs produced in a week. (a) Write the system of inequalities describing the constraints (including $x \geq 0$ and $y \geq 0$). (b) Find the vertices of the feasible region. (c) Determine the number of tables and chairs that maximizes the weekly profit $P = 70x + 30y$, and find the maximum profit.

(a) The constraints are $x \geq 0$, $y \geq 0$, $4x + y \leq 32$ (carpentry), and $2x + y \leq 20$ (finishing). (b) The boundary lines meet the axes and each other at $(0, 0)$, $(8, 0)$, $(6, 8)$, and $(0, 20)$ — these are the vertices of the feasible region. (c) Evaluating $P = 70x + 30y$: $(0, 0) \rightarrow 0$; $(8, 0) \rightarrow 560$; $(6, 8) \rightarrow 660$; $(0, 20) \rightarrow 600$. The maximum profit is 660 dollars, achieved by producing 6 tables and 8 chairs per week.

Multiple choice — easy

- 29 Solve $2^{2x-1} = 128$.

- a. 5
- b. 7
- c. 3
- d. 4 ✓

$128 = 2^7$, so $2x - 1 = 7$, giving $x = 4$.

30 Solve $3^{2x-1} = 3$.

a. 3

b. 0

c. 2

d. 1 ✓

$3 = 3^1$, so $2x - 1 = 1$, giving $x = 1$.

31 Solve $2^{2x-1} = 2$.

a. 0

b. 1 ✓

c. 3

d. 2

$2 = 2^1$, so $2x - 1 = 1$, giving $x = 1$.

32 Solve $2^{2x-1} = 2048$.

a. 7

b. 6 ✓

c. 5

d. 11

$2048 = 2^{11}$, so $2x - 1 = 11$, giving $x = 6$.

33 Solve $2^{2x-1} = 32$.

a. 5

b. 3 ✓

c. 2

d. 4

$32 = 2^5$, so $2x - 1 = 5$, giving $x = 3$.

34 Solve $3^{2x-1} = 19683$.

a. 4

b. 5 ✓

c. 9

d. 6

$19683 = 3^9$, so $2x - 1 = 9$, giving $x = 5$.

35 Solve $3^{2x-1} = 2187$.

a. 3

b. 4 ✓

c. 7

d. 5

$2187 = 3^7$, so $2x - 1 = 7$, giving $x = 4$.

36 Solve $3^{2x-1} = 27$.

a. 2 ✓

b. 4

c. 3

d. 1

$27 = 3^3$, so $2x - 1 = 3$, giving $x = 2$.

37 Solve $2^{2x-1} = 512$.

a. 6

b. 4

c. 9

d. 5 ✓

$512 = 2^9$, so $2x - 1 = 9$, giving $x = 5$.

38 Solve $7^{2x-1} = 16807$.

a. 4

b. 2

c. 5

d. 3 ✓

$16807 = 7^5$, so $2x - 1 = 5$, giving $x = 3$.

39 Solve $7^{2x-1} = 1977326743$.

a. 6 ✓

b. 11

c. 7

d. 5

$1977326743 = 7^{11}$, so $2x - 1 = 11$, giving $x = 6$.

40 Solve $7^{2x-1} = 40353607$.

a. 4

b. 6

c. 9

d. 5 ✓

$40353607 = 7^9$, so $2x - 1 = 9$, giving $x = 5$.

41 Solve $5^{2x-1} = 48828125$.

a. 11

b. 5

c. 6 ✓

d. 7

$48828125 = 5^{11}$, so $2x - 1 = 11$, giving $x = 6$.

42 Solve $3^{2x-1} = 243$.

a. 2

b. 3 ✓

c. 5

d. 4

$243 = 3^5$, so $2x - 1 = 5$, giving $x = 3$.

43 Solve $5^{2x-1} = 125$.

a. 1

b. 3

c. 4

d. 2 ✓

$125 = 5^3$, so $2x - 1 = 3$, giving $x = 2$.

Multiple choice — medium

44 Use the remainder theorem to find the remainder when $f(x) = x^3 - 5x^2 + 4x$ is divided by $(x - 0)$.

a. -2

b. 3

c. 2

d. 0 ✓

$f(0) = 0^3 = 0$. The remainder is 0 .

45 Use the remainder theorem to find the remainder when $f(x) = x^3 + 5x^2 - x - 5$ is divided by $(x - 1)$.

a. 0 ✓

b. -5

c. -2

d. 2

$f(1) = 1^3 + 5 - 1 - 5 = 0$. The remainder is 0 .

46 Use the remainder theorem to find the remainder when $f(x) = x^3 - 2x^2 - 5x - 1$ is divided by $(x - (-4))$.

a. -75

b. -79

c. -45

d. -77 ✓

$f(-4) = (-4)^3 - 32 + 20 - 1 = -77$. The remainder is -77 .

47 Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 - 2x + 2$ is divided by $(x - (-4))$.

a. -70 ✓

b. -54

c. -72

d. -68

$f(-4) = (-4)^3 - 16 + 8 + 2 = -70$. The remainder is -70 .

48 Use the remainder theorem to find the remainder when $f(x) = x^3 + 6x^2 + 9x - 7$ is divided by $(x - (-5))$.

a. -29

b. -25

c. -177

d. -27 ✓

$f(-5) = (-5)^3 + 150 - 45 - 7 = -27$. The remainder is -27 .

49 Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 3x - 5$ is divided by $(x - (-4))$.

a. -27

b. -23

c. -57

d. -25 ✓

$f(-4) = (-4)^3 + 32 + 12 - 5 = -25$. The remainder is -25 .

50 Use the remainder theorem to find the remainder when $f(x) = x^3 + x^2 - 7x + 2$ is divided by $(x - (-4))$.

- a. -16
- b. -18 ✓
- c. -34
- d. -20

$f(-4) = (-4)^3 + 16 + 28 + 2 = -18$. The remainder is -18.

51 Use the remainder theorem to find the remainder when $f(x) = x^3 + 5x^2 + 9$ is divided by $(x - 2)$.

- a. 37 ✓
- b. 35
- c. 17
- d. 39

$f(2) = 2^3 + 20 + 9 = 37$. The remainder is 37.

52 Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 + 7x - 9$ is divided by $(x - 5)$.

- a. 151
- b. 128
- c. 126 ✓
- d. 124

$f(5) = 5^3 - 25 + 35 - 9 = 126$. The remainder is 126.

53 Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 - 4x - 8$ is divided by $(x - (-2))$.

- a. -14
- b. -12 ✓
- c. -10
- d. -8

$f(-2) = (-2)^3 - 4 + 8 - 8 = -12$. The remainder is -12.

54 Use the remainder theorem to find the remainder when $f(x) = x^3 - 4x - 4$ is divided by $(x - (-1))$.

a. 1

b. 2

c. -3

d. -1 ✓

$f(-1) = (-1)^3 + 4 - 4 = -1$. The remainder is -1.

55 Use the remainder theorem to find the remainder when $f(x) = x^3 + 6x^2 - 4x - 7$ is divided by $(x - (-5))$.

a. 40

b. 38 ✓

c. -112

d. 36

$f(-5) = (-5)^3 + 150 + 20 - 7 = 38$. The remainder is 38.

56 Use the remainder theorem to find the remainder when $f(x) = x^3 - 3x^2 + 6x + 1$ is divided by $(x - (-1))$.

a. -7

b. -9 ✓

c. -11

d. -6

$f(-1) = (-1)^3 - 3 - 6 + 1 = -9$. The remainder is -9.

57 Use the remainder theorem to find the remainder when $f(x) = x^3 - 5x^2 + x - 4$ is divided by $(x - (-5))$.

a. -259 ✓

b. -261

c. -134

d. -257

$f(-5) = (-5)^3 - 125 - 5 - 4 = -259$. The remainder is -259 .

58 Use the remainder theorem to find the remainder when $f(x) = x^3 + 4x^2 + x + 7$ is divided by $(x - 0)$.

a. 10

b. 7 ✓

c. 9

d. 5

$f(0) = 0^3 + 7 = 7$. The remainder is 7.

Multiple choice — hard

59 Solve $\log_2(x + 3) + \log_2(x - 3) = 4$.

a. 7

b. 3

c. -5

d. 5 ✓

$\log_2[(x + 3)(x - 3)] = 4 \Rightarrow x^2 - 9 = 16 \Rightarrow x^2 = 25 \Rightarrow x = 5$ (rejecting the negative root since $x > 3$ is required).

60 Solve $\log_2 (x + 2) + \log_2 (x - 2) = 5$.

a. 6 ✓

b. 8

c. -6

d. 4

$\log_2 [(x + 2)(x - 2)] = 5 \Rightarrow x^2 - 4 = 32 \Rightarrow x^2 = 36 \Rightarrow x = 6$ (rejecting the negative root since $x > 2$ is required).

61 Solve $\log_2 (x + 1) + \log_2 (x - 1) = 3$.

a. -3

b. 5

c. 3 ✓

d. 1

$\log_2 [(x + 1)(x - 1)] = 3 \Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = 3$ (rejecting the negative root since $x > 1$ is required).

62 Solve $\log_3 (x + 3) + \log_3 (x - 3) = 3$.

a. 4

b. 6 ✓

c. -6

d. 8

$\log_3 [(x + 3)(x - 3)] = 3 \Rightarrow x^2 - 9 = 27 \Rightarrow x^2 = 36 \Rightarrow x = 6$ (rejecting the negative root since $x > 3$ is required).

63 Solve $\log_3(x + 4) + \log_3(x - 4) = 2$.

a. 7

b. -5

c. 3

d. 5 ✓

$\log_3[(x + 4)(x - 4)] = 2 \Rightarrow x^2 - 16 = 9 \Rightarrow x^2 = 25 \Rightarrow x = 5$ (rejecting the negative root since $x > 4$ is required).

64 Solve $3^x = 189$ for x , giving your answer to three decimal places.

a. 5.771

b. 5.271

c. 4.271

d. 4.771 ✓

$$x = \frac{\ln 189}{\ln 3} \approx 4.771.$$

65 Solve $5^x = 141$ for x , giving your answer to three decimal places.

a. 2.575

b. 4.075

c. 3.075 ✓

d. 3.575

$$x = \frac{\ln 141}{\ln 5} \approx 3.075.$$

66 Solve $7^x = 361$ for x , giving your answer to three decimal places.

a. 4.026

b. 2.526

c. 3.026 ✓

d. 3.526

$$x = \frac{\ln 361}{\ln 7} \approx 3.026.$$

67 Solve $5^x = 372$ for x , giving your answer to three decimal places.

a. 3.178

b. 4.178

c. 4.678

d. 3.678 ✓

$$x = \frac{\ln 372}{\ln 5} \approx 3.678.$$

68 Solve $5^x = 410$ for x , giving your answer to three decimal places.

a. 3.738 ✓

b. 4.238

c. 4.738

d. 3.238

$$x = \frac{\ln 410}{\ln 5} \approx 3.738.$$

69 Solve the system $y = x^2 + 4x + 1$ and $y = -5x - 7$. Find the x -coordinates of the intersection points.

a. $-8, -3$

b. $-8, -1$ ✓

c. $-6, -1$

d. $8, 1$

$$x^2 + 4x + 1 = -5x - 7 \Rightarrow x^2 + 9x + 8 = 0 \Rightarrow (x - (-8))(x - (-1)) = 0 \Rightarrow x = -8 \text{ or } x = -1.$$

70 Solve the system $y = x^2 + x + 10$ and $y = -5x + 10$. Find the x-coordinates of the intersection points.

a. $-4, 0$

b. $6, 0$

c. $-6, 0$ ✓

d. $-6, -2$

$$x^2 + x + 10 = -5x + 10 \Rightarrow x^2 + 6x = 0 \Rightarrow (x - (-6))(x - 0) = 0 \Rightarrow x = -6 \text{ or } x = 0.$$

71 Solve the system $y = x^2 + 17x + 47$ and $y = 2x - 9$. Find the x-coordinates of the intersection points.

a. $-8, -9$

b. $-8, -7$ ✓

c. $8, 7$

d. $-6, -7$

$$x^2 + 17x + 47 = 2x - 9 \Rightarrow x^2 + 15x + 56 = 0 \Rightarrow (x - (-8))(x - (-7)) = 0 \Rightarrow x = -8 \text{ or } x = -7.$$

72 Solve the system $y = x^2 + 13x + 27$ and $y = 4x + 7$. Find the x-coordinates of the intersection points.

a. $-5, -4$ ✓

b. $5, 4$

c. $-3, -4$

d. $-5, -6$

$$x^2 + 13x + 27 = 4x + 7 \Rightarrow x^2 + 9x + 20 = 0 \Rightarrow (x - (-5))(x - (-4)) = 0 \Rightarrow x = -5 \text{ or } x = -4.$$

73 Solve the system $y = x^2 - 8x + 16$ and $y = 3x - 8$. Find the x -coordinates of the intersection points.

a. 3, 8 ✓

b. 5, 8

c. 3, 6

d. -3, -8

$$x^2 - 8x + 16 = 3x - 8 \Rightarrow x^2 - 11x + 24 = 0 \Rightarrow (x - 3)(x - 8) = 0 \Rightarrow x = 3 \text{ or } x = 8.$$